



Bop or Flop: Predicting Popularity of Taylor Swift Songs, and What Makes Them Bops?

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Purpose of Study

The music industry is routinely referred to as one of the most competitive professions (Keyes, October 2021). Since the field is so competitive, it can be very challenging to be successful and create a song that will stand out from the crowd. An estimated 90% of music artists fail and never become successful (Keyes, October 2021). There are many different reasons that these musicians are not able to make it in such a competitive industry. Factors not even related to music such as advertising and release schedules can impact success regardless of the quality of their music (Keyes, October 2021).

Taylor Swift is clearly not struggling as she is leading the music industry right now. According to Time, Swift was the top streaming artist on both Spotify and Apple Music in 2023. On top of that, she also has the highest-grossing world tour, The Eras Tour. Swift's tour has had an estimated \$5.7 billion boost to the United States economy (McCluskey, December 2023). It is clear that Swift has found success in such a competitive industry. She has an incredibly diverse discography with many different tracks comprised of a vast array of styles. By analyzing what makes Swift so popular at a deeper level it may be possible for new artists entering the industry to emulate her style and try to enter the mainstream music industry.

Given that, what sets Taylor Swift apart from others in the industry? We set out to identify predictors that lead to the vast popularity of Taylor Swift songs. This could be used by Swift herself and other similar artists.

Data Source, Details, and Drawbacks

The dataset used for this analysis is from Kaggle named "Taylor Swift Spotify Dataset" (Priester, April 2024). The Kaggle user, Jarred Priester, has curated and updated this spreadsheet when needed. Not having the ability to see the data collection process can make it difficult to understand the data, but insight was gained by looking into Priester's past projects. His relevant past projects include ones such as, "Predicting Sign Language Letters using a Convolutional Neural Network", "Unsupervised Recommendation System of The Beatles", and "Heart Disease Predictions using a ML Ensemble in R" (Priester, J., May 2024). With these in mind, it can be assumed this dataset was also created using some variation of a machine learning model.

The original dataset directly from Kaggle contains 530 observations with 17 variables. This includes 27 albums that were a mix of original releases, deluxe editions, "Taylor's Version" recordings, and live recordings. Taylor Swift has chosen to re-record many of her earlier albums in an attempt to reclaim ownership of her art as the masters of her original albums were sold against her wishes (Grady, 2023).

The decision was made to focus on the original releases of Swift's albums and to not include the "Taylor's Version" re-recordings of her first six albums. This is because the re-recordings have performed very well on the Spotify streaming platform in comparison to the original album releases. If the re-recordings were included, the results of this modeling effort would have been heavily skewed. Therefore, the dataset consists of songs from her 10 original album releases.

Thus, the final dataset used for analysis has 172 observations that span 10 of Swift's albums ("Taylor Swift" to "Midnights"). The most recent release, The Tortured Poets Department: The Anthology (April 2024) is not included. Shown in Table 1 are the original names and their corresponding description of the predictor variables used in this study. All these predictors are numerical and continuous and are from the Kaggle dataset cited above.

Table 1. Summary Table of Predictor Variables

Predictor Variable	Description
Acousticness	Confidence measure of the song being acoustic. 0 to 1, 1 represents high confidence of being acoustic.
Danceability	0: least danceable, 1: most danceable
Energy	0 to 1, measure of intensity and activity
Instrumentalness	Predicts whether a song has no vocals. 1 represents greater likelihood of containing no vocal content.
Liveness	Detects the presence of an audience in the recording. 0 to 1, higher values represent increased chance that the track was performed live.
Loudness	Loudness of track in decibels. Values typically range from -60 to 0 db.
Speechiness	Detects the presence of spoken words in a track. 0 to 1, 1 represents an exclusively speech-like track.
Tempo	Overall estimated tempo of a track in beats per minute (BPM).
Valence	Describes the musical positiveness conveyed by a track. 0 to 1, 0 represents a sad or angry song, 1 represents a very positive and happy song
Duration_ms	Length of song in milliseconds

Two response variables were used for this assessment (Table 2). The first response variable, popularity, is continuous and was taken from the original dataset. The second response variable, popularity_binary, was created from the continuous response variable, popularity. This was done as it is necessary to have a binary response variable for some of the methods used in this study. The median of the original response variable, popularity (69) was used as the threshold for the binary response variables – if a song had a popularity value less than 69 it would be assigned a 0 and 69 or greater would be 1. Essentially, 0 is a “flop” while 1 is a “bop”. Table 2 summarizes these response variables:

Table 2. Summary of Response Variables

Response Variables	Description
Popularity	Measurement of 0-100, our dataset had values from 40-98
Popularity (binary)	Divided dataset at median value of popularity (69), 0 it is a flop, 1 it is a bop

Additional variables in the original data set, such as name (the song name), album, and track_number were not used for modeling but were useful for exploratory data analysis and identifying specific observations. Specifically, the song name was used as an observation identifier. The variables completely removed from the dataset were id and uri as these are Spotify-exclusive values that would not help with modeling.

Some of the major drawbacks of the data used in this study are as follows:

- 1) This dataset was exclusively curated with statistics from Spotify. Information from other streaming platforms or physical media were not included; thus, if they were included the results of this effort could vary.
- 2) The modelers and authors of this effort were not the ones to create the dataset or machine learning model used to create the dataset.
- 3) The real-world context of the songs is also not included (i.e., which songs were picked as singles, had a music video, had a sound trend on social media, etc.).

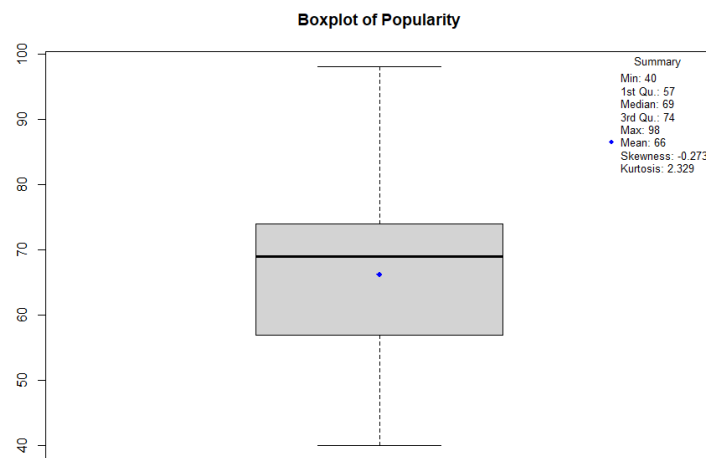
Exploratory Data Analysis

The data analysis for this project was conducted using the R language while utilizing various packages to facilitate the modeling process. RStudio was employed as the integrated development environment, providing a comprehensive platform for scripting, debugging, and visualizing the dataset.

As previously discussed, with the reduction in the overall dataset via removing the “Taylor Version” re-records of Swift’s previous albums, the remaining dataset contained 172 observations. Among the 10 predictors being used, there were no missing values for any of the observations/songs; therefore, no imputations nor dropping of additional observations were necessary. The original “popularity” variable was maintained as a response variable. A binary version of this variable was created by classifying songs into a “flop” class and “bop” class based on whether or not they had a popularity value greater than 69 (the median of the popularity variable).

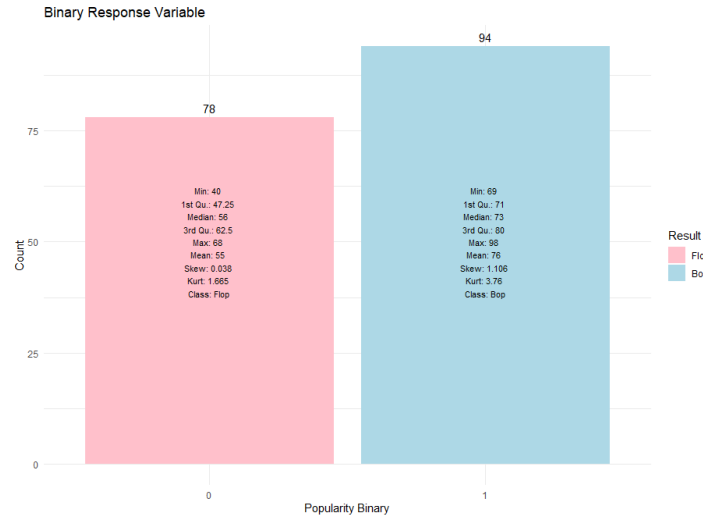
The boxplot below (Figure 1) shows that the popularity response variable does not contain any outliers and has a very slight left skewness to it. The range of the popularity response variable is 58.

Figure 1. Boxplot of Response Variable - Popularity



The classes for the binary response variable do not appear to have any alarming skewness or kurtosis to them. The “bop” category has slightly more observations, but overall, the balance between the two classes is unalarming (Figure 2).

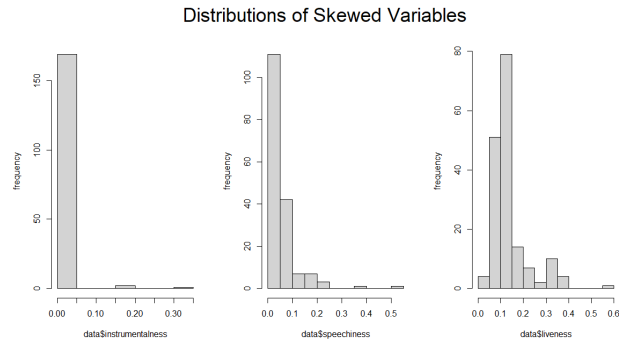
Figure 2. Bar Plot Statistical Summary of Binary Response Variable



The kurtosis and skewness of each of the predictor variables was then assessed, as some of the models to be developed will require these predictor variables to be normally distributed. Through examining Table 3, it appears that out of the ten predictor variables, instrumentality, liveness, and speechiness had the most extreme values of skewness and kurtosis. Looking at histograms of these variables, it was apparent they needed to be transformed.

Table/Figure 3. Skewness and Kurtosis Analysis of Predictor Variables

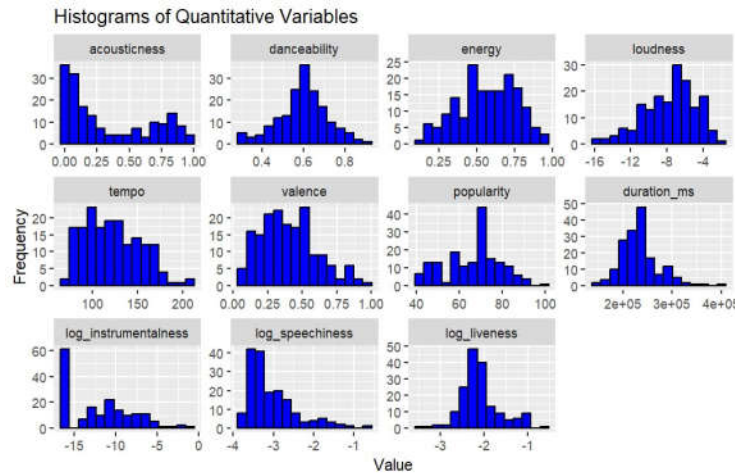
Variable	Skewness	Kurtosis
acousticness	0.6220680	1.789459
danceability	-0.2561319	3.109043
energy	-0.1948785	2.300508
instrumentality	8.7009479	83.770061
liveness	2.3278154	9.787388
loudness	-0.5506356	2.878547
speechiness	4.3043051	27.778909
tempo	0.4265068	2.468030
valence	0.4803181	2.754774
duration_ms	1.0123600	5.125985



As a result, these variables were all natural log transformed, and the 0 values for instrumentality were transformed to 1E-07 (an order of magnitude lower than the lowest non-zero value for instrumentality). A table showing the reduction of the skewness and kurtosis that remains after this transformation for these three variables is included in the Appendix. The analysis of skewness and kurtosis were done using the e1071 R package (Meyer D., Dimitriadou E., Hornik K., Weingessel A., Leisch F., 2023)

Histograms of each of the quantitative variables (including the response variable, popularity) were created to assess their overall distribution (Figure 4). The only remaining variables that could potentially be problematic for normality assumptions are acousticness and duration_ms; however, they were left as is.

Figure 4. Histograms of Quantitative Variables



The predictor variables were also assessed for outliers via Tukey fences, and six of them (danceability, loudness, duration_ms, and the three recently transformed variables) had outliers as seen in boxplots included in the Appendix. Note no other observations were dropped from the dataset to mitigate these outliers. The predictors were also assessed for outliers per the binary response classes as well. Eight out of the ten predictors had some outliers, with energy and tempo having no outliers when the data was split into the binary response classes. Again, no observations were dropped from the dataset to mitigate these outliers. These boxplots are also included in the Appendix.

A correlation matrix was also developed to assess for any collinearity among the predictors (Figure 5). Energy and loudness appear to have a positive correlation with each other (0.78) and acousticness appears to have a negative correlation energy (-0.66) and loudness(-0.73). Three multiple linear regression models were run with all the data and other predictors, with each of these models having one of these potential interaction terms. Table 4 lists the model, the interaction term tested, the estimate and standard error of said estimate for each term, and a p-value. The only interaction term shown to be significant was energy:loudness. As a result, the interaction term of energy:loudness was added to the dataset as an 11th predictor variable.

Figure 5. Correlation Matrix of Predictor and Response Variables

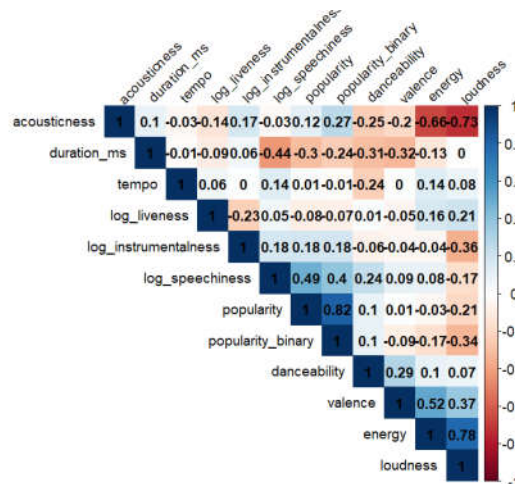


Table 4. Summary of Check for Interaction Terms

	Model	Interaction	Estimate	StdError	Pr
1	lm.fit1	acousticness:energy	2.471713	15.617941	8.74e-01
2	lm.fit2	acousticness:loudness	1.016263	1.178507	3.90e-01
3	lm.fit3	energy:loudness	-3.197443	1.469538	3.10e-02

Each predictor variable was also individually assessed to see if there was any non-linear relationship between the predictor and the original popularity response variable. This assessment was done from the second to the eighth power. Table 5 below summarizes the p-values from that assessment. Boxplots used to aid in this part of the assessment are also included in the Appendix.

Table 5. Assessment of Polynomial Relationships – Predictor vs. Response

	Predictor	P_B2	P_B3	P_B4	P_B5	P_B6	P_B7	P_B8
1	acousticness	0.6140	0.4645	0.3701	0.3117	0.2745	0.2503	0.2345
2	danceability	0.1250	0.1086	0.0941	0.0816	0.0709	0.0618	0.0541
3	energy	0.1592	0.1680	0.1708	0.1704	0.1690	0.1677	0.1673
4	loudness	0.6619	0.5475	0.4534	0.3781	0.3186	0.2716	0.2343
5	tempo	0.0691	0.0751	0.0808	0.0858	0.0900	0.0934	0.0959
6	valence	0.8129	0.8497	0.9118	0.9775	0.9619	0.9087	0.8628
7	duration_ms	0.8154	0.8058	0.7971	0.7894	0.7825	0.7765	0.7714
8	log_instrumentalness	0.1981	0.1521	0.1229	0.1042	0.0921	0.0845	0.0799
9	log_speechiness	0.1497	0.1467	0.1477	0.1521	0.1595	0.1696	0.1821
10	log_liveness	0.7278	0.7049	0.6812	0.6579	0.6357	0.6149	0.5961
11	energy_loudness	0.9445	0.9786	0.9899	0.9620	0.9382	0.9183	0.9021

Assuming an $\alpha = 0.05$, it appears that none of the predictors have a non-linear relationship with the continuous popularity response variable. Of note, some of the predictors do have p -values between 0.05 and 0.10 (see tempo, danceability, log_instrumentalness); however, the exploration of more complex polynomials, splines, and generalized additive models will not be further explored in this assessment.

Methods Used for Modeling and Validation

As previously discussed, there are two response variables to be assessed: the popularity variable, which is continuous, and the binary popularity variable created via splitting the dataset into two by using the median of the popularity variable (69) as a cutoff for whether or a song is a “flop” (below 69) or a bop (above 69). Therefore, Table 6 summarizes the methods used per the response variable attempting to be predicted:

Table 6. Methods Used in This Study

Response Variable: Popularity	Response Variable: Binary Popularity
Multiple Linear Regression - Basic - Best Subset – Exhaustive - Shrinkage: Ridge and Lasso - Principle Component/Partial Least Squares	Logistic Regression Linear Discriminant Analysis Naïve Bayes Classification Trees Support Vector Classifiers/Machines
Regression Trees: - Bagging - Random Forest - Boosting	- Linear - Radial - Polynomial

For assessing training and test mean square error for models using the continuous response variable and for assessing error rates for variables using the binary response variable, we randomly split the data where 50% of the observations ended up in a training dataset to train the

respective models and 50% of the observations ended up in a testing dataset to assess the predictive ability of the resultant models.

Modeling Methods for Continuous Response Variable (popularity)

The first methods addressed in this section are variations of multiple linear regression that involve variable selection, coefficient shrinkage, of component creation. Regression tree methods and their results will wrap up this section

Variable or Component Reduction/Minimization

The following sections review methods used to adjust the multiple linear regression model training process via selecting the best variables per certain criteria (such as adjusted R^2 , C_p , and BIC), reducing or selecting coefficients via a penalty, or through developing new components that are linear combinations of the original variables in hopes of reducing predictor dimensionality in a final model.

Multiple Linear Regression - Basic

The basic multiple linear regression assessment was initiated via running a basic multiple linear regression with the 11 predictor variables on the training dataset using the `lm` R function from the stats package. The output from R that summarizes the result of this fit is shown in Table 7 below. Out of the 11 predictor variables, only two (`log_speechiness` with a p -value of 0.0134 and the interaction term of `energy:loudness` with a p -value of 0.0852) ended up as being either as significant or borderline significant if one were to assume an $\alpha = 0.05$. Additionally, the F -statistics of 3.077 and the associated p -value of 0.001903 indicate that there is a relationship between the predictors and the response variable (popularity).

Table 7. R Output of Basic Multiple Linear Regression Model

```
Call:
lm(formula = popularity ~ ., data = updated_data,
    subset = train)

Residuals:
    Min       1Q   Median       3Q      Max
-21.093   -8.209    1.536    6.750   23.446

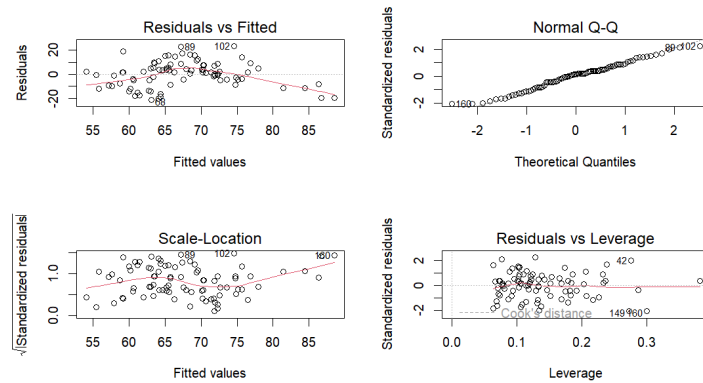
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.148e+02  2.494e+01   4.605 1.68e-05 ***
acousticness  3.275e+00  5.983e+00   0.547  0.5857
danceability  6.295e+00  1.177e+01   0.535  0.5943
energy       -2.646e+01  2.097e+01  -1.262  0.2110
loudness      1.301e+00  1.359e+00   0.957  0.3417
tempo       -4.198e-02  4.241e-02  -0.990  0.3254
valence     -3.147e+00  7.935e+00  -0.397  0.6928
duration_ms  -4.469e-05  4.286e-05  -1.043  0.3005
log_instrumentalness 4.929e-01  3.576e-01   1.378  0.1723
log_speechiness  6.726e+00  2.654e+00   2.534  0.0134 *
log_liveness   -4.862e-01  2.908e+00  -0.167  0.8677
energy_loudness -3.850e+00  2.207e+00  -1.744  0.0852 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.27 on 74 degrees of freedom
Multiple R-squared:  0.3139,    Adjusted R-squared:  0.2119
F-statistic: 3.077 on 11 and 74 DF,  p-value: 0.001903
```

A Shapiro-Wilk normality test was completed on the residuals of this model using the `shapiro.test` stats package function in R. The null hypothesis of this test is that the residuals of the model are normally distributed, and the alternative is that the residuals of the model are not normally distributed. The resulting p -value of that test was 0.2842, thus the residuals of this multiple linear regression are normally distributed.

The residuals versus fitted plot, a Normal Q-Q plot, a Scale-Location plot, and a Residuals versus Leverage plot were also created and evaluated to see how the model is performing on a more granular level and to see if there were any outliers identified (see Figure 6).

Figure 6. Graphical Analysis of Residuals for Multiple Linear Regression



There were a handful of songs in the training dataset identified as outliers in the plots above. Those songs were “Enchanted”, “Don’t Blame Me”, “Cruel Summer”, “Closure”, and “Vigilante [expletive]”. For the Residual vs. Fitted plot, it appears the model may overestimate the extremes of the fitted values as this can be observed via the trendline of the residuals being below the 0 residual line at lower fitted popularity values and at higher fitted popularity values. For the Scale-Location plot it appears that the points are randomly dispersed around a horizontal line, however, there appears to be a slight dip in the mid-range of the popularity values and a slight increase at higher values; however, it appears that the assumption of homoscedasticity is likely met. For the Residuals vs. Leverage plot, none of the points are outside the Cook’s Distance contour of 0.5, therefore none of the observations are singularly impacting the model’s coefficients, and most of the data points are clustered around the horizontal zero axis with standardized residual values from -2 to 2, thus indicating most of the residuals are within two standard deviations of the predicted values.

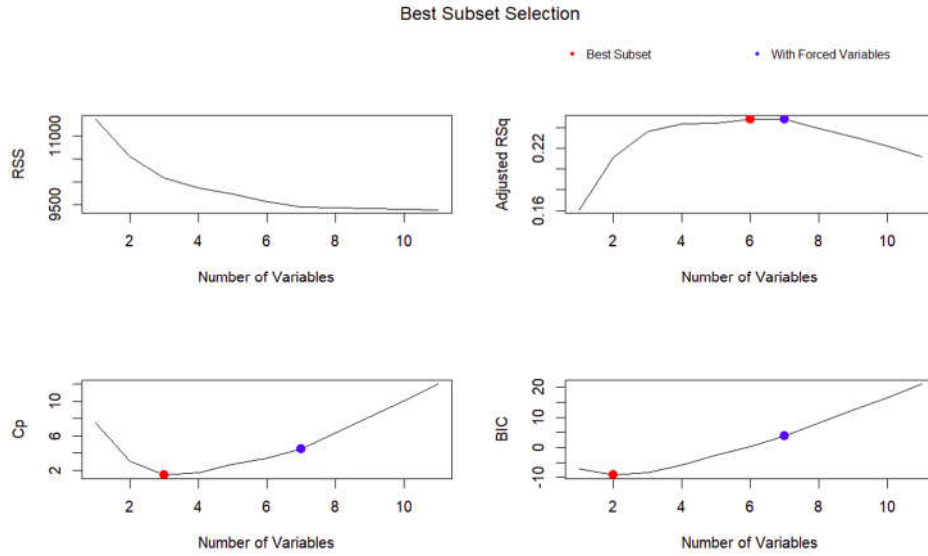
This fitted model was then used to predict the response variable value using both the training data set and the test data set. The resulting mean square errors for these two sets were 109.2344 and 129.4887, respectively.

Best Subset Selection

An exhaustive search was done when attempting to reduce the number of variables in the model via Best Subset Selection. This process entails evaluating all possible models for each number of predictors up to a certain max. The R function used to do this, `regsubsets` in the `leaps` package (Miller, A., 2020), does this for up to 8 predictors, but if a dataset has more than that number of predictors, it uses a “branch and bound” algorithm that is also effectively exhaustive but more efficient (Lumley, T., 2020). This fitting was done using the training dataset

Figure 7 shows plots of how RSS , adjusted R^2 , C_p , and BIC change with the addition of more variables to the model.

Figure 7. Plots for Best Subset Selection Method



The model with the highest adjusted R^2 value had 6 variables, the model with the lowest C_p value had 3 variables, and the variable with the lowest BIC value had 2 variables (the red dots on the graphs). Using the adjusted R^2 variable count cutoff leads to a model where the interaction term, energy:loudness, is included, but only the main effect energy is in the model. Therefore, the best subset selection model had to be forced to 7 variables to include the main effect of loudness (see blue dots on Figure 7). Table 8 below captures the variables included in this model, their respective coefficients, and whether they were significant.

Table 8. Summary Table of Best Subset Selection

Variable	Coefficient	p-value
energy*	-3.1487e+01	0.09556
loudness	1.2344e+00	0.32535
tempo	-4.3378e-02	0.28750
duration_ms	-4.7059e-05	0.23106
log_instrumentalness	4.8228e-01	0.16457
log_speechiness**	6.6689e+00	0.00982
energy:loudness*	-4.0550e+00	0.05707

* -Significant

An attempt to do forward and backward subset selection was also done, but the same result came through due to the nature of forcing the main effects of the interaction term into the model.

The residuals of this model created using the training data were also normally distributed per the Shapiro-Wilk residual test (p -value = 0.1163), and the residual plots and outliers were very similar to the basic multiple linear regression model that was previously discussed.

This fitted model was then used to predict the response variable value using both the training data set and the test data set. The resulting mean square errors for these two sets were 109.9215 and 130.18, respectively.

Shrinkage Methods – Ridge and Lasso

As an alternative to the subset selection discussed above, techniques to constrain or regularize the coefficient estimates can be used to shrink the coefficient near to zero. These methods are referred to as the ridge and lasso methods. A shrinkage penalty for these two methods is paired with a tuning parameter, λ , to control the relative impact of the shrinkage penalty. The shrinkage penalty for the ridge method is $\sum_{j=1}^p \beta_j^2$ and the shrinkage penalty for the lasso method is $\sum |\beta_j|$. (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023b).

A model using the ridge penalty was developed using the glmnet R function (Friedman J, Tibshirani R, Hastie T, 2010), the training data set, and a grid of lambdas ranging from 1E10 to 1E-2. The same grid, R function, and training data set were also used for developing a model using the lasso penalty via a minor adjustment to the alpha parameter in the function. All the variables should be standardized prior to implementing these methods for fitting a multiple linear regression. The glmnet R function automatically standardizes predictor variables before fitting the model and also centers the response variable. It then returns the coefficients to the original scale by rescaling them after the model fit (Friedman, J., Hastie, T., and Tibshirani, R., 2020). Specifically for the lasso method, during the initial run of the method the interaction term energy:loudness came through with a non-zero coefficient, but the main effect loudness did not/its coefficient was shrunk to zero. The method was rerun using a penalty factor label on the other variables to allow for the loudness variable to come through. This is a built-in feature of the glmnet R function that can be utilized in scenarios such as this (Friedman, J., Hastie, T., and Tibshirani, R., 2020).

Figures showing the choice of the best lambda for each method via cross-validation completed by the cv.glmnet R function, the coefficients plotted over several values of lambda, and the coefficients plotted over either $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$ for the ridge regression or $\|\hat{\beta}_\lambda^R\|_1 / \|\hat{\beta}\|_1$ for the lambda regression are included in the appendix to this report. The best lambda for the ridge regression was 11.22 and the best lambda for the lasso regression was 0.44.

Table 9 summarizes the resulting coefficients from each method. The coefficients of the ridge regression were not dramatically different from those observed in the basic multiple linear regression, with the exception of energy, loudness, and valence being reduced by almost an order of magnitude. The lasso regression shrunk the coefficient of acousticness, valence, and log_liveness to zero. The coefficient for loudness was also shrunk significantly by the lasso method. These fitted models were then used to obtain a mean square error for the training and test dataset. The resulting mean square errors for the ridge regression model for the training and test sets were 116.7989 and 134.8741, respectively, and 111.8964 and 130.6318 for the lasso regression model, respectively.

Table 9. Summary of Ridge and Lasso Regression Results - Coefficients

Variable	Basic Multiple Linear Regression Coefficients	Ridge Regression Coefficients	Lasso Regression Coefficients
acousticness	3.275e+00	1.7621e+00	.
danceability	6.295e+00	5.6575e+00	2.9566e+00
energy	-2.646e+01	-3.4530e+00	-1.5450e+01
loudness	1.301e+00	-3.2851e-01	4.1500e-02
tempo	-4.198e-02	-2.7412e-02	-2.8934e-02
valence	-3.147e+00	-5.7519e-01	.
duration_ms	-4.469e-05	-3.2721e-05	-3.14703e-05
log_instrumentalness	4.929e-01	2.6803e-01	2.8859e-01
log_speechiness	6.726e+00	3.6098e+00	5.900e+00
log_liveness	-4.8623e-01	-3.2110e-01	.
energy:loudness	-3.850e+00	-1.0998e+00	-2.2158e+00

Principal Component Regression (PCR) and Partial Least Squares Regression (PLS)

Instead of the actual variables (or slightly transformed variables), PCR and PLS identify linear combinations of the original features and replace the variables with a small set of principal components that capture the joint variation. Then those small set of components are used to fit a linear model using ordinary least squares. The components identified are in order of highest variance to lowest variance and are orthogonal to each other. A major difference between PCR and PLS is that PLS makes use of the response variable when developing the components, but PCR does not. The direction of the components in PCR are identified in an unsupervised manner where the response variable is not used to direct the direction of the principal components. As a result, PCR does not guarantee that components that best explain predictors will be best for predicting response (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023b). When evaluating these models, one wants to identify the model with the smallest cross-validation error as well as the model that is adequate at explaining the variance in the predictors as well as the response variable.

For the PCR model, the `pcr` R function from the `pls` library was used. The function can be used to scale the predictors prior to generating components, so that the scale of which each variable does not have an effect. This function also computes ten-fold cross validation error for each of the possible number of principle components used (Mevik, B.H. and Wehrens, R., May 2023).

Figures 8 and 9 present the validation plot of the cross-validated root mean square error prediction (RMSEP) values and scree plots examining the cumulative percentage explained in the predictors and response variables, respectively. Looking at Figure 8, the lowest cross-validation error is observed at two components; however, Figure 9 shows that the explanation of the variance of the predictors does not reach 90% until 7 components and only reaches roughly 30% explanation of variance in the response variable by 7 components. As a result, the model

with 7 components was chosen to predict the response using the training and testing data sets. The resulting MSEs for those two sets were 112.5988 and 134.4601.

Figure 8. Validation Plot of PCR Method

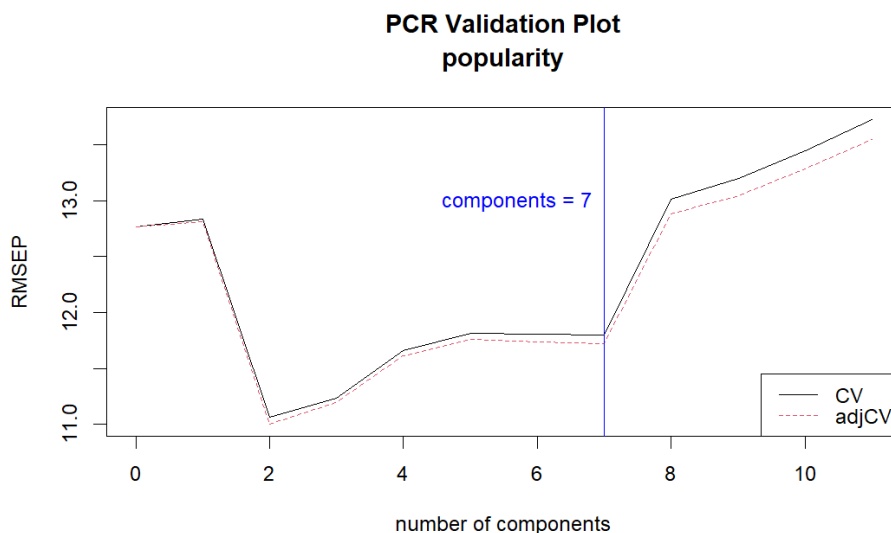
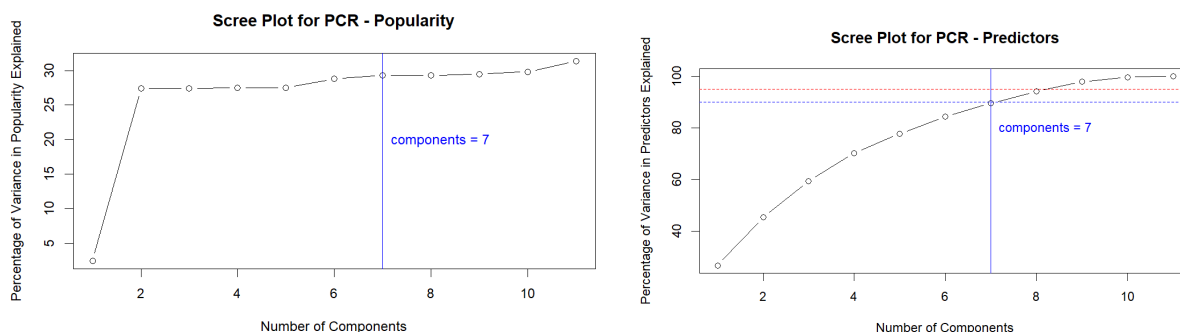


Figure 9. Scree Plots of Percent Variance Explained of Predictors and Response Variables - PCR



For the PLS model, the `plsR` R function from the `pls` library was used. Like the `pcr` function, this function can be used to scale the predictors prior to generating components, so that the scale of which each variable does not have an effect. This function also computes ten-fold cross validation error for each of the possible number of principle components used (Mevik, B.H. and Wehrens, R., May 2023).

Figures 10 and 11 on the next page present the validation plot of the cross-validated RMSEP values and scree plots examining the cumulative percentage explained in the predictors and response variables, respectively. Looking at Figure 10, the lowest cross-validation error is observed at one component; however, Figure 11 shows that the explanation of the variance of the predictors does not reach 90% until 9 components and only reaches roughly 30% explanation of variance in the response variable by 9 components. As a result, the model with 9 components was chosen to predict the response using the training and testing data sets. The resulting MSEs for those two sets were 109.2349 and 129.5703.

Figure 10. Validation Plot of PLS Method

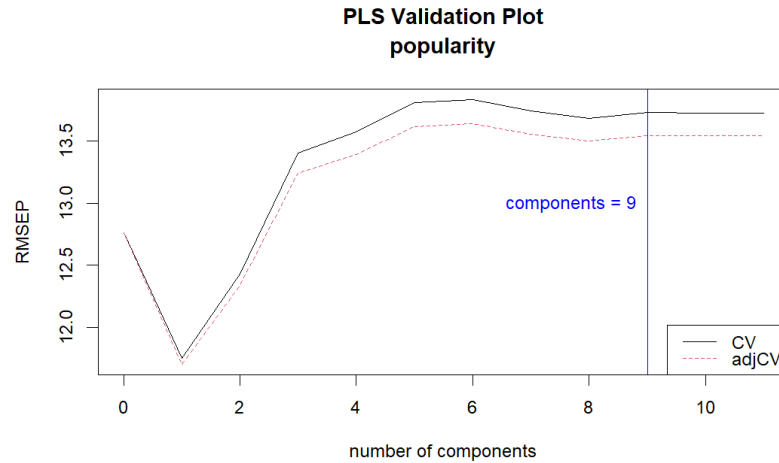
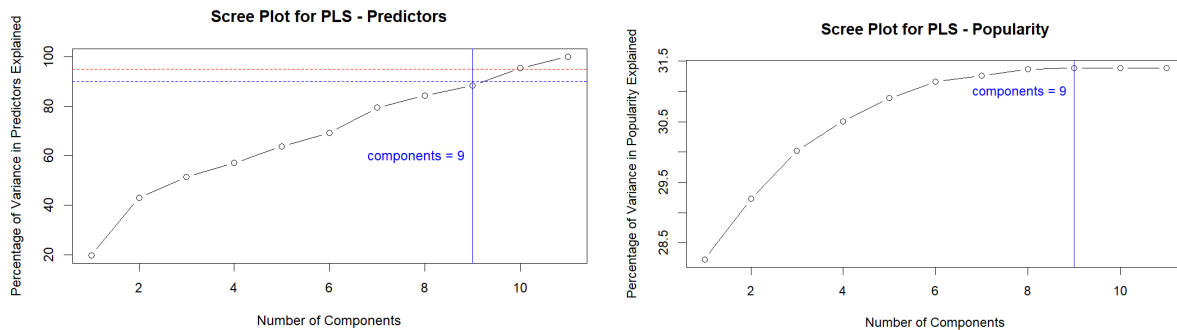


Figure 11. Scree Plots - Percent Variance Explained of Predictors and Response Variables (PLS)



Summary of Multiple Linear Regression Methods

Table 10 summarizes the MSE values obtained for the training data set and for the test data set for each of the multiple linear regression methods explored in this project. The lowest MSE was observed with the basic multiple linear regression, however PLS came in a close second. The results of the PLS model are not surprising as a dramatic reduction in dimensionality was not observed (i.e., the basic multiple linear regression model has 11 predictor variables, while the PLS model only reduced the dimensionality down to 9 components).

The limitations of the lasso regression and best subset regression models was the need to force main effect variables into the model due to the interaction term of energy:loudness. If that interaction term was not significant, it is possible more efficient models would have been created with these two methods.

The PCR model and ridge regression performed similarly. The PCR model is near the bottom of this list of performance likely due to the response variable not being considered when the direction of the components are identified. The ridge regression also only moderately shrunk the coefficients from the original basic multiple linear regression.

In summation, the basic multiple linear regression model performed the best out of the methods used to reduce/minimize variables and components.

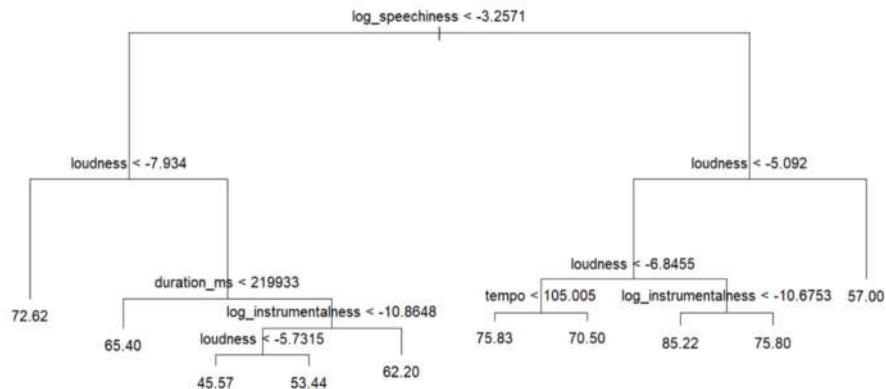
Table 10. Summary of Multiple Linear Ridge Regression

	Model	Train_Error	Test_Error
1	Least Squares	109.2344	129.4887
6	PLS	109.2349	129.5703
2	Best Subset/F/B	109.9215	130.1800
4	Lasso	111.8964	130.6318
5	PCR	112.5988	134.4601
3	Ridge Regression	116.7989	134.8741

Regression Trees

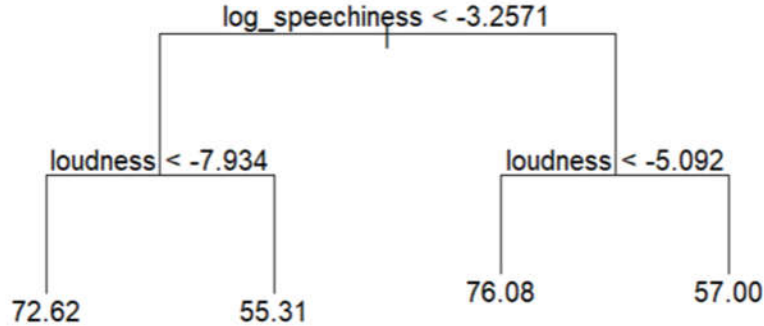
For the initial regression tree analysis, the means of training observations from the same region as the given test observation are utilized to predict its response. It is essential to initially examine the complete tree before pruning down to the most relevant branches (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023c). This comprehensive view is generated using the `tree()` function from the R package `tree` (Ripley, B., 2023). The model incorporates 11 predictors from a subset of the training data, and the full regression tree is depicted in Figure 12. Analysis of the internal node hierarchy reveals that `log_speechiness` and `loudness` are the most influential predictors in this model. The fully expanded tree comprises 10 leaves, which are used for making predictions. Performance metrics for this model include a training Mean Squared Error (MSE) of 36.588 and a testing MSE of 161.106.

Figure 12. Regression Tree Before Pruning



Pruning tree models is crucial to reduce the residual sum of squares. This process involves creating a smaller, optimal subtree from the initial larger one (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023c). The optimal size of this subtree is found through cross-validation using the `cv.tree()` function in R (Ripley, B., 2023). The optimal size corresponds to the lowest cross-validation error rate. For this regression tree, the optimal size was four leaves with a cross-validation error rate of 10958.29. Figure 13 shows the tree result of refitting the tree with four leaves using `prune.tree()` function. Based on the resulting three branches, `log_speechiness` and `loudness` were the most influential, as they were in the initial un-pruned tree. The train MSE for the pruned tree was 67.179 and the test MSE was 153.589. This results in a slightly better test MSE than the full tree.

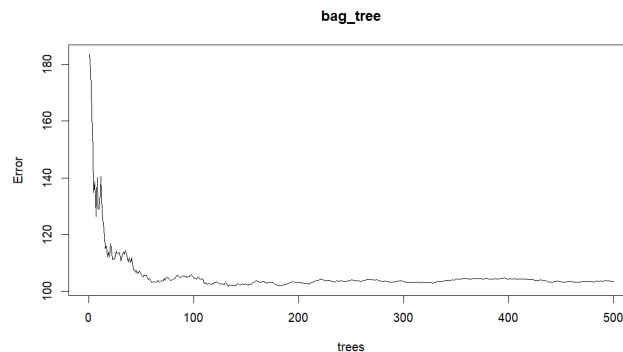
Figure 13. Regression Tree After Pruning



Bagging

Bagging is a technique used to reduce the variance in tree regression models. It accomplishes this by constructing B regression trees, each based on a bootstrapped training set, and then averaging the prediction of these trees. Each tree created from this method has a high variance and a low bias (bias-variance tradeoff), but the process of averaging these trees significantly reduces this variance (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023c). Bagging is performed using the `randomForest()` function from the `randomForest` library in R (Liaw, A. and Wiener, M., 2002). The parameter `mtry` is set to 11 signifying to use 11 variables (predictors) to randomly sample as candidates at each split. This results in a train MSE of 17.641 and a test MSE of 122.543. 500 trees are created when running this process. Figure 14 shows how as the number of trees increases, the error goes down.

Figure 14. Bagging Plot – Number of Trees Versus Error



Random Forest

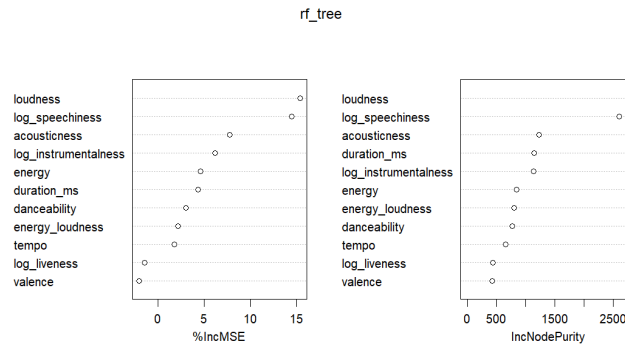
The random forest method, which is similar to bagging, uses the `randomForest()` function to create a model (Liaw, A. and Wiener, M., 2002). The primary distinction between random forests and bagged trees lies in the selection of the predictor subset (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023c). For this random forest model, the `mtry` parameter was set to five, allowing an analysis of which variables significantly impact the prediction of popularity. This will help show which variables are important for predicting popularity. Table 11 and Figure 15 present the importance table and variable importance plot for the random forest model, respectively. Both indicate that `loudness` and `log_speechiness` are the most important variables, consistent with the findings from the initial and pruned regression tree. The training MSE is

18.596 and the test MSE is 118.856. The testing MSE shows an improvement when using the random forest model to subset the model, thus highlighting its effectiveness in enhancing predictive accuracy.

Table 11 Importance Table of Random Forest Model

	%IncMSE	IncNodePurity
acousticness	7.767188	1232.6259
danceability	3.010608	770.0786
energy	4.636683	848.3192
log_instrumentalness	6.163980	1132.9779
log_liveness	-1.425596	438.5824
loudness	15.394148	2774.4997
log_speechiness	14.480703	2604.3562
tempo	1.782163	658.2772
valence	-2.030601	428.0594
duration_ms	4.361311	1151.1106
energy_loudness	2.198164	804.2179

Figure 15. Variable Importance Plot for Random Forest Model



Boosting

Boosting involves fitting a generalized boosted regression model where trees are grown sequentially. Each new tree is developed using information from the previous tree to continuously refine the model (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023b). This model is performed using the `gbm()` function from the `gbm` library in R (Greg, R., Developers G, 2024). The numeric results are shown in Table 12, which shows that the most important predictors are `log_speechiness`, `loudness`, `log_instrumentalness`, `energy`, and `energy_loudness`. While more predictors are included in the results, `log_speechiness` and `loudness` remain the most influential, consistent with other tree methods explored thus far. For boosting, the train MSE is 61.786 and the test MSE is 139.752. This is a slight increase for the test MSE which is not ideal, but boosting was helpful to see which predictors are also important.

Table 12. Boosting - Relative Influence Table

	var	rel.inf
log_speechiness	log_speechiness	19.994564
loudness	loudness	16.802391
log_instrumentalness	log_instrumentalness	11.648371
energy	energy	11.484764
energy_loudness	energy_loudness	10.188047
duration_ms	duration_ms	7.006863
tempo	tempo	6.166624
valence	valence	5.780916
danceability	danceability	4.424823
acousticness	acousticness	4.031021
log_liveness	log_liveness	2.471617

Modeling Methods for Binary Response Variable (popularity_binary)

The following subsections go over the modeling methods used to predict the popularity_binary response variable. The response variable, popularity_binary, has two classes where 0 indicates that the song is a flop and 1 that the song is a bop.

Logistic Regression

Logistic regression employs the maximum likelihood method to fit a model (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023a). In R, this is performed using the glm() function, which is part of the default stats package and is used to fit generalized linear models. By setting the family parameter in the function to binomial, it specifies that the model should be treated as a logistic regression. Assuming an $\alpha = 0.05$, the significant predictors are acousticness, energy, loudness, log_speechiness, and energy_loudness. These are highlighted in Table 13.

Table 13. Output of Logistic Regression Model

```
Call:
glm(formula = popularity_binary ~ acousticness + danceability +
    energy + log_instrumentalness + log_liveness + loudness +
    log_speechiness + tempo + valence + duration_ms + energy_loudness,
    family = binomial, data = data, subset = train)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.8484  -0.5757   0.2128   0.7435   1.4732

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.550e+01  7.400e+00  2.095  0.03618 *
acousticness  3.491e+00  1.537e+00  2.271  0.02317 *
danceability  4.127e+00  3.149e+00  1.310  0.19008
energy       -1.216e+01  5.465e+00 -2.225  0.02609 *
log_instrumentalness 5.729e-02  8.404e-02  0.682  0.49543
log_liveness  4.852e-02  6.675e-01  0.073  0.94205
loudness      9.041e-01  3.664e-01  2.467  0.01361 *
log_speechiness 2.571e+00  1.031e+00  2.493  0.01267 *
tempo        -1.604e-02  1.176e-02 -1.363  0.17277
valence      -2.725e+00  2.054e+00 -1.327  0.18462
duration_ms   2.373e-07  9.859e-06  0.024  0.98080
energy_loudness -1.812e+00  6.221e-01 -2.913  0.00358 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 116.227  on 85  degrees of freedom
Residual deviance:  75.025  on 74  degrees of freedom
AIC: 99.025

Number of Fisher Scoring iterations: 6
```

The confusion matrices for the training and test data are shown below in Table 14. For both the training and test data, there is a decent accuracy rate. There is also a relatively low error rate for each.

Table 14. Confusion Matrices for Training and Test Data – Logistic Regression

Training Data - Logistic Regression						Test Data - Logistic Regression					
Predict	Class	Truth				Predict	Class	Truth			
		0	1					0	1		
	0	28	3	NPV	0.90		0	25	8	NPV	0.76
	1	7	48	Precision	0.87		1	18	35	Precision	0.66
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.80	0.94	0.884	0.116			0.58	0.81	0.698	0.302

Linear Discriminant Analysis

Linear Discriminant Analysis assumes a multivariate gaussian distribution for each predictor, implying that each follows a normal distribution with some degree of correlation between pairs

(James, G., Witten, D., Hastie, T., Tibshirani, R., 2023a). These assumptions were verified in the exploratory data analysis. This analysis was performed using the `lda()` function from the MASS library (Venables, W. N., Ripley, B. D., 2002). The prior probabilities for class 0 are 0.4069767 and class 1 is 0.5930233. This means the probabilities of a song being a hit is more likely than it being labeled a flop. Table 15 displays the coefficients of the linear discriminants, with the most significant predictors being those whose coefficients have the largest absolute value (Brownlee, J., 2020). For this model, the most significant predictors for predicting `popularity_binary` are `acousticness`, `energy`, `loudness`, `log_speechiness`, and `energy_loudness`. A figure showing the distribution of the probability of each song for each class is included in the Appendix.

Table 15. Coefficients of Linear Discriminants

Coefficients of Linear Discriminants		LD1
acousticness		2.0580
danceability		2.5699
energy		-7.0988
log_instrumentalness		0.0522
log_liveness		0.0820
loudness		0.5185
log_speechiness		0.9231
tempo		- 0.0065
valence		-1.7691
duration_ms		-1.4805e-6
energy_loudness		-1.1344

The confusion matrices for the training and test data are shown below in Table 16. For both the training and test data, there is a fairly decent accuracy rate. There is also a relatively low error rate for each.

Table 16. Confusion Matrices for Training and Test Data – LDA

Training Data - LDA						Test Data - LDA					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	27	7	NPV	0.79		0	24	7	NPV	0.77
	1	8	44	Precision	0.85		1	19	36	Precision	0.65
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.77	0.86	0.826	0.174			0.56	0.84	0.698	0.302

Naïve Bayes

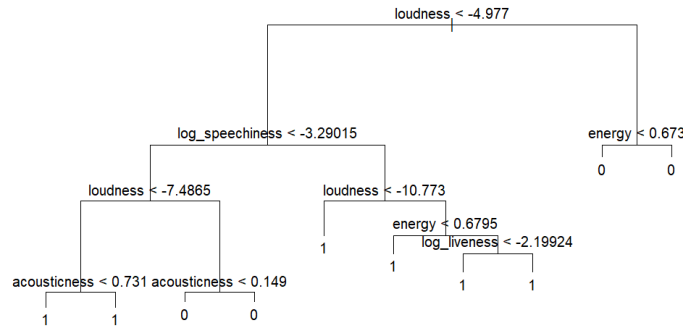
This method utilizes Bayes' Theorem to estimate π_k as the proportion of training observations belonging to the k th class (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023a). In R, this is done using the `naiveBayes()` function from the `e1071` library (Meyer D., Dimitriadou E., Hornik K., Weingessel A., Leisch F., 2023). The full output of the test is in the Appendix. The largest differences between the two classes are `log_instrumentalness`, `log_liveness`, `loudness`, `log_speechiness`, `tempo`, `duration_ms` and `energy_loudness`. Table 17 presents the confusion matrices for the training and test data. The error rates were very similar to LDA.

Table 17. Confusion Matrices for Training and Test Data – Naïve Bayes

Training Data - Naïve Bayes						Test Data - Naïve Bayes					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	29	10	NPV	0.74		0	25	8	NPV	0.76
	1	6	41	Precision	0.87		1	18	35	Precision	0.66
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.83	0.80	0.814	0.186			0.58	0.81	0.698	0.302

Classification Trees

For the classification tree, each observation is predicted to belong to the most commonly occurring class of training observations in the region to which it is assigned (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023c). Much like the regression tree, the full model is initially fit. This is done using the `tree()` function in R (Ripley, B., 2023). The full model fit is shown in Figure 17 with nine branches and ten leaves. Below that, Table 18 presents the confusion matrices for the training and test data. The accuracies are reasonable, but it is important to prune the tree to potentially improve these values and reduce the classification error.

Figure 17. Classification Tree Before Pruning**Table 18. Confusion Matrices for Training and Test Data – Class Tree Pre-Prune**

Training Data - Class Tree Pre Prune						Test Data - Class Tree Pre Prune					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	27	2	NPV	0.93		0	24	3	NPV	0.89
	1	8	49	Precision	0.86		1	19	40	Precision	0.68
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.77	0.96	0.884	0.116			0.56	0.93	0.744	0.256

Pruning for classification trees is conducted through cross-validation `cv.tree()` (Ripley, B., 2023). After running this function, it was determined that the optimal number of leaves is 4. This corresponds to the minimum cross-validation error rate of 23. The resulting tree is shown in Figure 18. The confusion matrices shown below in Table 19 indicate that pruning had little impact, particularly with the test data. Consequently, both the pre-prune and pruned trees do the same job of predicting the response variable, popularity_binary.

Figure 18. Classification Tree After Pruning

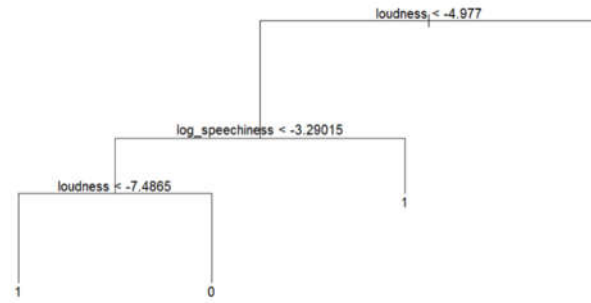


Table 19. Confusion Matrices for Training and Test Data – Class Tree Post-Prune

Training Data - Class Tree Post Prune						Test Data - Class Tree Post Prune					
Predict	Class	Truth				Predict	Class	Truth			
	0	24	3	NPV	0.89		0	24	3	NPV	0.89
	1	19	40	Precision	0.68		1	19	40	Precision	0.68
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.56	0.93	0.744	0.256			0.56	0.93	0.744	0.256

Support Vector Classifiers/Machines

The binary population target was used as well as the 10 predictor variables and one interaction term to create predictive models using support vector machines/classifiers. Three different kernels were explored: linear, radial, and polynomial. When the linear kernel is used a support vector classifier is fitted, and when a radial or polynomial kernels are used a support vector machine is fitted (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023d). As done with the other models in the assessment, fitted models were trained with a training dataset and their accuracy tested with a test dataset. These training and test datasets are the same ones used throughout this entire effort. Once the initial support vector classifier or machine was fitted, the parameters key to the respective kernel were tuned using the training data. The tune function of the e1071 R package was used to complete the tuning, which uses 10-fold cross-validation on a set of models of interest (Meyer D., Dimitriadou E., Hornik K., Weingessel A., Leisch F., 2023). The resulting tuned model was also tested using the test data set.

Linear Kernel

For the linear kernel, the cost parameter was tuned among cost values ranging from 0.01 to 10. A larger cost value would result in narrower margins and there will be few support vectors on the margin or violating the margin, a smaller cost value would result in the opposite (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023d). Initially the support vector classifier was fit with a cost value of 0.01. The resulting number of support vectors was 51, with 25 being in the flop class and 26 being in the bop class. The resulting training and test error/accuracy rates are shown below in Table 20.

Table 20. Confusion Matrices for Training and Test Data – Linear Kernel Pre-Tune

Training Data – Pre-Tune: Linear						Test Data – Pre-Tune: Linear					
Predict	Class	Truth				Predict	Class	Truth			
	0	11	8	NPV	0.58		0	8	7	NPV	0.53
	1	24	43	Precision	0.64		1	35	36	Precision	0.51
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.31	0.84	0.63	0.37			0.19	0.84	0.51	0.49

This model was then tuned to a cost value of 0.05, which resulted in a narrower margin. This resulted in 62 support vectors with the vectors being equally split between the two classes and a marked decrease in the error rates and improvement in the accuracy, precision, specificity, and sensitivity rates. Table 21 summarizes the resulting training and test error/accuracy rates for the tuned model.

Table 21. Confusion Matrices for Training and Test Data – Linear Kernel Tuned

Training Data – Tuned: Linear						Test Data – Tuned: Linear					
Predict	Class	Truth				Predict	Class	Truth			
		0	1					0	1		
	0	26	2	NPV	0.93		0	20	3	NPV	0.87
	1	9	45	Precision	0.83		1	23	40	Precision	0.63
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.74	0.96	0.87	0.13			0.47	0.93	0.70	0.30

Radial Kernel

Moving onto the support vector machines, the radial kernel was initially fitted using a cost value of 0.05 and a gamma of 0.0909 (which is equivalent to 1 divided by the predictors being used in the model). In comparison to the adjustment of the cost value for the support vector classifier, the adjustment in gamma during the tuning process will also have impacts on the decision boundary. A larger gamma value, the reach of the support vectors becomes very short, meaning the model can capture finer nuances of the data, potentially leading a model to fit training data very closely and resulting in a decision boundary that is more irregular. Conversely, a low gamma broadens the reach of support vectors, leading to a decision boundary with lower variance that can generalize better, but may underfit complex data (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023d).

This fit resulted in 86 support vectors (i.e., all the observations in the training set) with 35 being in the flop class and 51 being in the bop class. The resulting training and test error/accuracy rates are shown below in Table 22.

Table 22. Confusion Matrices for Training and Test Data – Radial Kernel Pre-Tune

Training Data - Pre-Tune: Radial						Test Data - Pre-Tune: Radial					
Predict	Class	Truth				Predict	Class	Truth			
		0	1					0	1		
	0	0	0	NPV	-		0	0	0	NPV	-
	1	35	51	Precision	0.59		1	43	43	Precision	0.50
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.00	1.00	0.59	0.41			0.00	1.00	0.50	0.50

One can see that the pre-tuned radial kernel performed poorly and did not classify a single song into the flop class

The radial kernel model was tuned using cost values ranging from 0.01 and 10 and gamma was also tuned using values ranging from 0.01 to 10. The tuned cost increased to 0.5 and the gamma decreased to a value of 0.05. The resulting tuned model had 69 support vectors, with 33 being in the flop class and 36 being in the bop class. The resulting training and test error/accuracy rates are shown in Table 23 and show a marked improvement with tuned parameters.

Table 23. Confusion Matrices for Training and Test Data – Radial Kernel Tuned

Training Data - Tuned: Radial						Test Data - Tuned: Radial					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	28	3	NPV	0.90		0	20	3	NPV	0.87
	1	7	48	Precision	0.87		1	23	40	Precision	0.63
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.80	0.94	0.88	0.12			0.47	0.93	0.70	0.30

Polynomial Kernel

The third kernel assessed was the polynomial kernel. The pre-tune cost was set to the 0.5 value reached with the tuning of the radial kernel, and the gamma was similarly set to 0.05. The degree value was also set to 2, which would be indicative of a quadratic boundary (James, G., Witten, D., Hastie, T., Tibshirani, R., 2023d). The initial fit resulted in 47 support vectors with 24 being in the flop class and 23 being in bop class. The resulting training and test error/accuracy rates are shown below in Table 24, and are by far the worst out of the three kernels in the pre-tuning stage, but unlike the radial kernel the polynomial kernel did predict some flops:

Table 24. Confusion Matrices for Training and Test Data – Polynomial Kernel Pre-Tune

Training Data - Pre-Tune: Polynomial						Test Data - Pre-Tune: Polynomial					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	8	18	NPV	0.31		0	9	19	NPV	0.32
	1	27	33	Precision	0.55		1	34	24	Precision	0.41
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.23	0.65	0.48	0.52			0.21	0.56	0.38	0.62

The polynomial kernel model was tuned using cost values ranging from 0.01 and 10, gamma values ranging from 0.01 to 10, and degree values of 1, 2, 3, 4, and 5. The tuned cost increased to 5 and the gamma further decreased to a value of 0.01. The resulting tuned degree was 1, which is indicative of a linear boundary. The resulting tuned model had 62 support vectors, with 31 being in the flop class and 31 being in the bop class. The resulting training and test error/accuracy rates are shown below in Table 25 and show a marked improvement with tuned parameters:

Table 25. Confusion Matrices for Training and Test Data – Radial Kernel Tuned

Training Data - Tuned: Polynomial						Test Data - Tuned: Polynomial					
		Truth						Truth			
Predict	Class	0	1			Predict	Class	0	1		
	0	28	3	NPV	0.90		0	20	3	NPV	0.87
	1	7	48	Precision	0.87		1	23	40	Precision	0.63
		Specificity	Sensitivity	Accuracy	Error			Specificity	Sensitivity	Accuracy	Error
		0.80	0.94	0.88	0.12			0.47	0.93	0.70	0.30

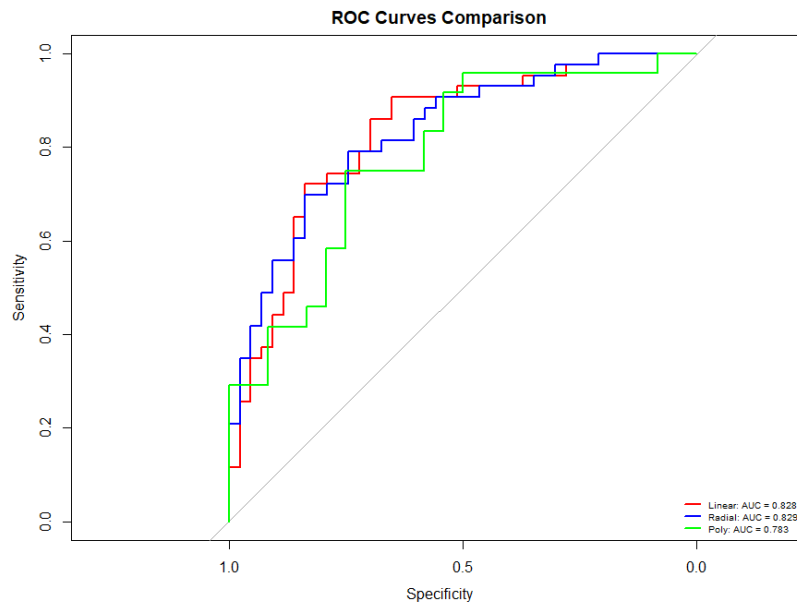
Summary of Support Vector Classifiers/Machines Results

Table 26 summarizes the various rates (accuracy, error, specificity, etc.) for all three models pre and post tuning:

Table 26. Summary of Confusion Matrix Calculations for SVM Models

Kernel	Tune	Accuracy		Error		Specificity		Sensitivity		NPV		Precision	
		Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
Linear	Pre	0.63	0.51	0.37	0.49	0.31	0.19	0.84	0.84	0.58	0.53	0.64	0.51
	Post	0.87	0.70	0.13	0.30	0.74	0.47	0.96	0.93	0.93	0.87	0.83	0.63
Radial	Pre	0.59	0.50	0.41	0.50	0.00	0.00	1.00	1.00	-	-	0.59	0.50
	Post	0.88	0.70	0.12	0.30	0.80	0.47	0.94	0.93	0.90	0.87	0.87	0.63
Poly	Pre	0.48	0.38	0.52	0.62	0.23	0.21	0.65	0.56	0.31	0.32	0.55	0.41
	Post	0.88	0.70	0.12	0.30	0.80	0.47	0.94	0.93	0.90	0.87	0.87	0.63

The resulting training and test errors post-tuning for all three models were nearly the same. The linear model appears to differ a little bit from the radial and polynomial, but not dramatically so. As a result, an ROC curve comparison was completed to try to determine if one of these models was slightly better than the other in terms of performance. Figure 19 shows these curves as well as their respective areas under them. It appears that the radial kernel model has a slightly larger area under the curve than the linear model.

Figure 19. SVM ROC Curves Comparison

Summary of Results

Table 28 below summarizes the test MSE or error rate for each of the models developed in this effort. For the continuous response variable (popularity), the regression trees using random forest (MSE = 11.856) and bagging (MSE = 122.543) outperformed the other regression trees and even the basic multiple linear regression method with the lowest test MSE, the basic multiple linear regression (MSE = 129.4887).

For the models created to predict the binary response variables (popularity_binary), the test errors were all identical except for the classification tree (error rate = 0.256). The similarity of the test errors for the binary response variable may be connected to the response variable arbitrarily being created from the continuous variable with a cutoff of the median of said variable.

Each of the models were also reviewed to see if predictor variables were identified as significant or important in the fit and function of the model. For the models created to predict the continuous response variable, predictors log_speechiness, energy, loudness, and the interaction term energy:loudness were the most consistently significant/important predictors. For the models created to predict the binary response variable the predictors acousticness, energy, loudness, log_speechiness, and the interaction term energy:loudness were significant/important.

It appears that energy, loudness, speechiness, and on occasion acousticness, are the key predictors (and interaction term) when predicting the popularity of Ms. Swift's music. Considering log_speechiness was one of the variables, perhaps she should pursue creating a rap album.

Table 28. Summary Table of Methods and Test MSE/Test Error Rates

Response Variable	Method:	Test MSE or Error Rate:
Popularity	Basic MLR	129.4887
	Ridge Regression	134.8714
	Lasso Regression	130.6318
	Best Subset - Exhaustive	130.1800
	PCR	134.4601
	PLS	129.5703
	Regression Tree - Basic	Pre-Prune: 161.106, Post-Prune: 153.589
	Regression Tree - Bagging	122.543
	Regression Tree – Random Forest	118.856
	Regression Tree - Boosting	139.752
Popularity_Binary	Logistic Regression	0.302
	LDA	0.302
	Naïve Bayes	0.302
	Classification Tree	Pre-Prune: 0.256, Post-Prune: 0.256
	Support Vector – Linear, Radial, Poly (all Post-Tune)	0.302

Conclusion

Model types that do the best job of predicting their respective response targets (popularity and popularity_binary) have been identified in this study. In the process of identifying those models, the key predictor variables out of the original set of predictor variables were also identified.

Table 29 below succinctly summarizes these results.

Table 29. Summary of Best Models and Significant/Important Predictors

Models:	Variables:
Continuous Response:	Energy
- Decision Tree-Random Forest - Decision Tree-Bagging - Basic MLR	Loudness
Binary Response:	Energy:Loudness
Classification Tree	Log_Speechiness
	Acousticness

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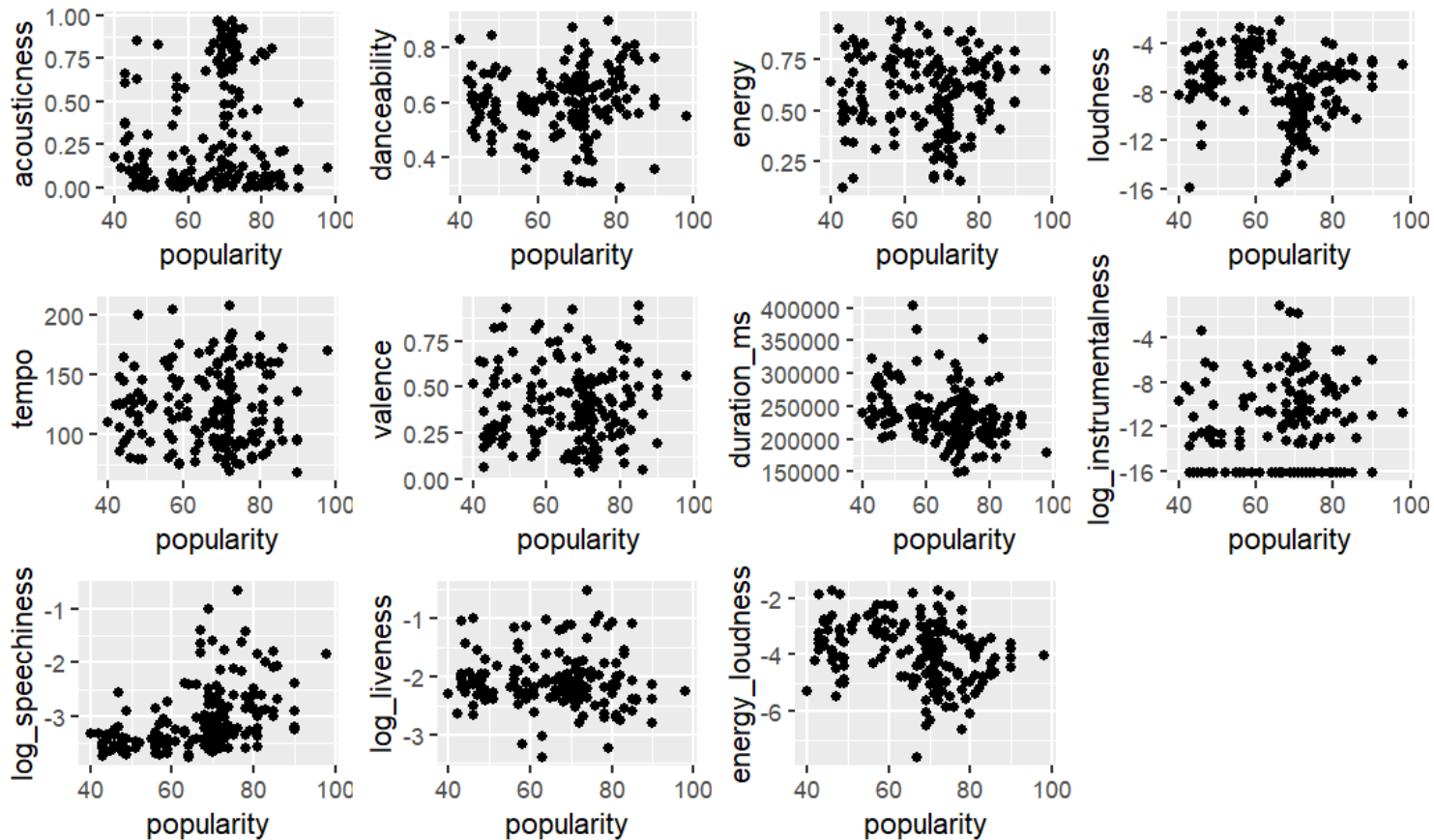
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APPENDIX

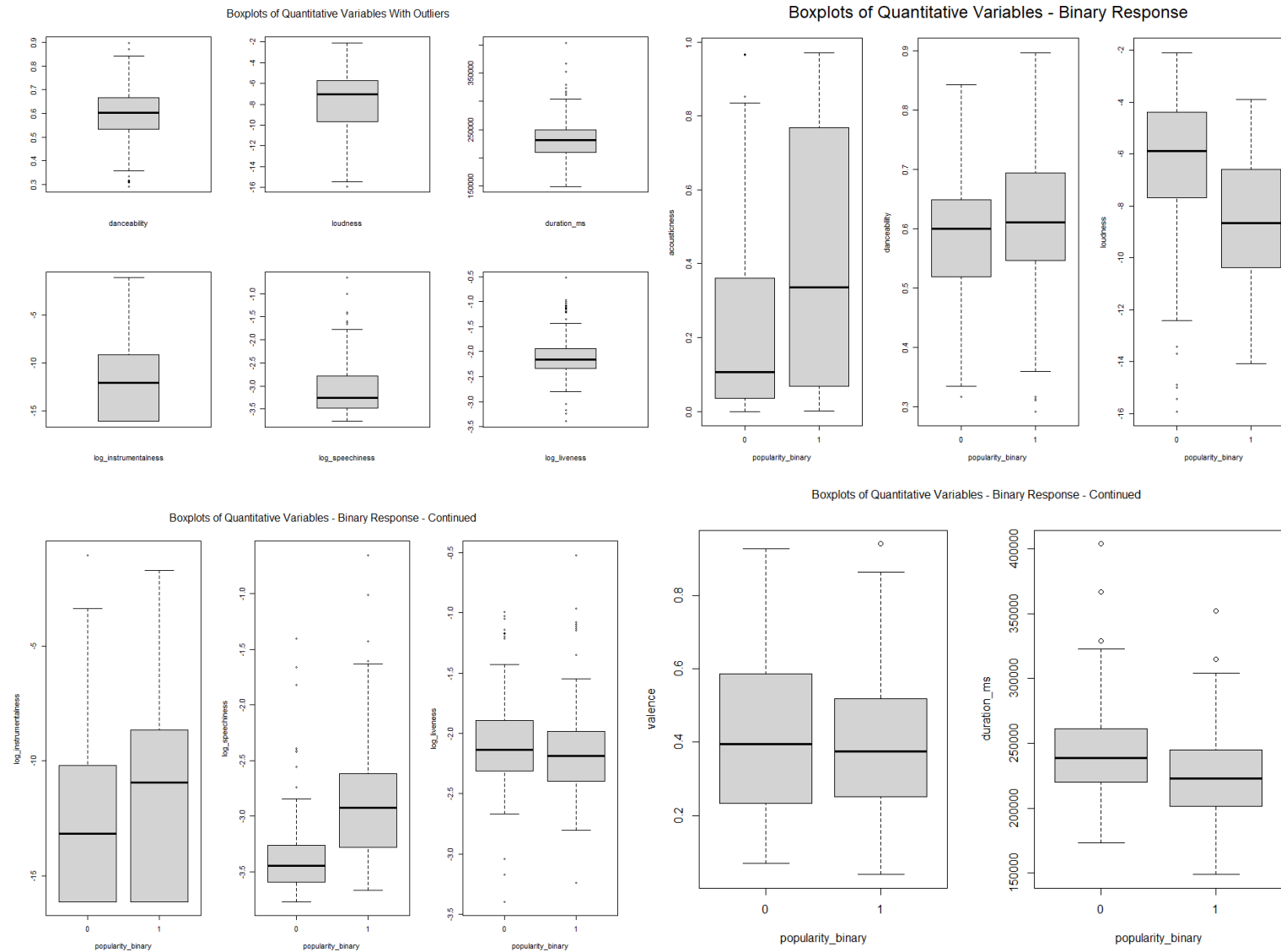
Skewness and Kurtosis of Transformed Predictor Variables

Variable	Skewness	Kurtosis
log_instrumentalness	0.4939175	2.384841
log_speechiness	1.4365396	4.991420
log_liveness	0.6875061	4.158902

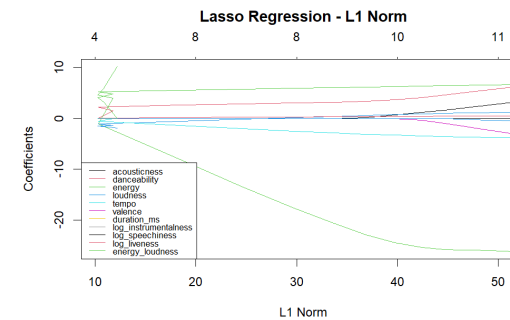
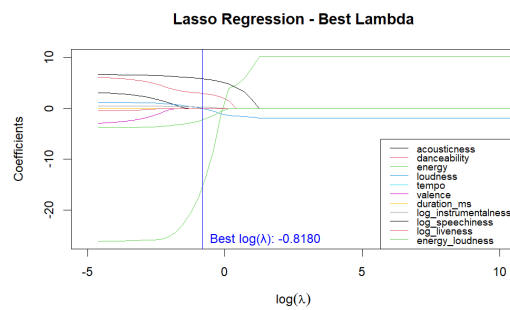
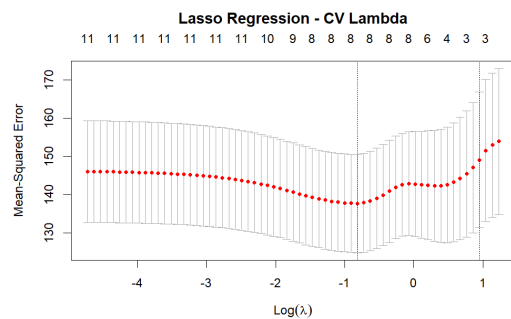
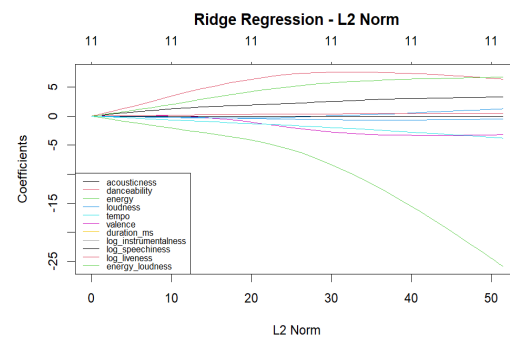
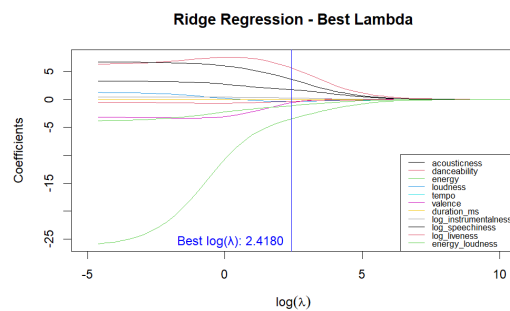
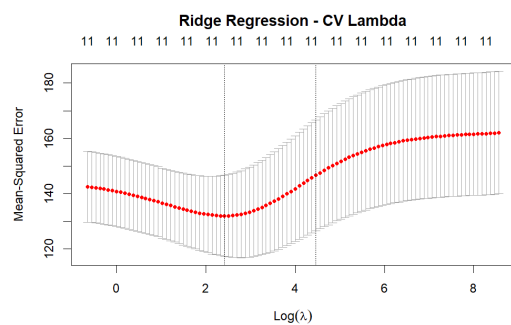
Scatter Plots of Quantitative Variables vs. Popularity Response (checking for non-linearity)



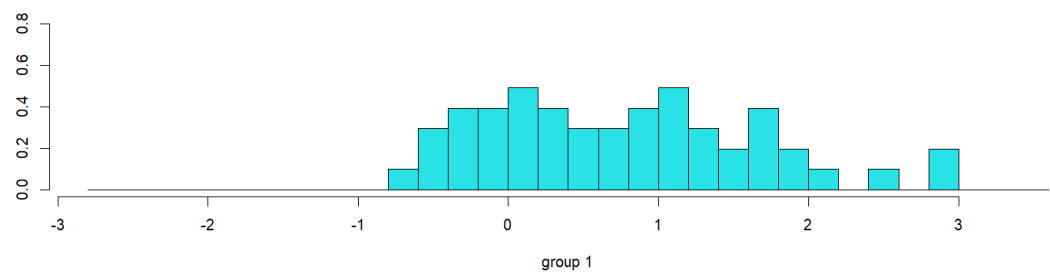
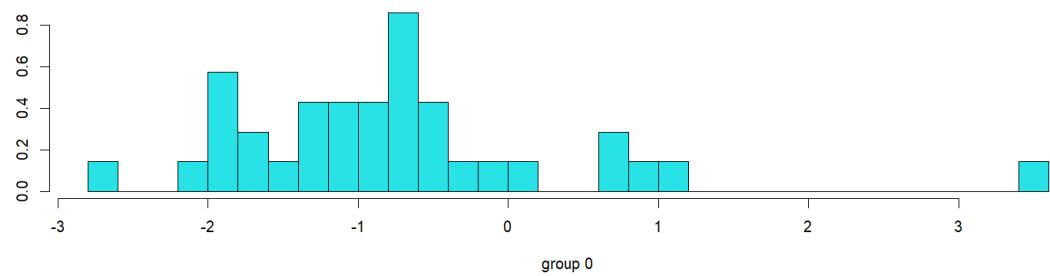
Boxplots from EDA showing outliers in predictor variables and in terms of the binary response variable:



Plots for Ridge and Lasso Regressions



Plot from LDA Analysis:



R Output for Naïve Bayes

Naïve Bayes Classifier for Discrete Predictors

Call:

```
naiveBayes.default(x = X, y = Y, laplace = laplace)
```

A-priori probabilities:

```
Y
  0      1
0.4069767 0.5930233
```

Conditional probabilities:

```
acousticness
Y      [,1]      [,2]
0 0.2274706 0.2860268
1 0.4422695 0.3454017
```

```
danceability
Y      [,1]      [,2]
0 0.5628286 0.1088705
1 0.6018627 0.1472757
```

```
energy
Y      [,1]      [,2]
0 0.6278000 0.2002898
1 0.5404902 0.1813153
```

```
log_instrumentalness
Y      [,1]      [,2]
0 -12.50182 3.928591
1 -11.03496 3.882633
```

```
log_liveness
Y      [,1]      [,2]
0 -2.140201 0.5114085
1 -2.147014 0.4497577
```

```
loudness
Y      [,1]      [,2]
0 -6.381771 3.331374
1 -8.288941 2.160079
```

```
log_speechiness
Y      [,1]      [,2]
0 -3.259210 0.4504952
1 -2.895943 0.6148116
```

```
tempo
Y      [,1]      [,2]
0 129.8330 28.45592
1 117.4122 32.36761
```

```
valence
Y      [,1]      [,2]
0 0.4432857 0.2303066
1 0.4066059 0.2071054
```

```
duration_ms
Y      [,1]      [,2]
0 239329.5 31323.76
1 228191.0 38795.32
```

```
energy_loudness
Y      [,1]      [,2]
0 -3.516305 1.065481
1 -4.194965 1.107669
```